VU Quiz Collection

Total Quizzes: 10

Courses: MTH401 - Differential Equations (Quiz 2), MTH603 - Numerical Analysis (Quiz No 02)

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Waseem Akram | wasii.dev | 03137119351

Quiz 1 - MTH401 - Differential Equations (Quiz 2)

Question:

IfA(t)istheamountofaradioactivesubstancepresentattimet,therateofchangeofA(t)withrespecttotisgivenby_____. IfA(t)istheamountofaradioactivesubstancepresentattimet,therateofchangeofA(t)withrespecttotisgivenby_____. \begin{gathered}
{\text{If}}\,A(t)\,\,{\text{is}}\,\,{\text{the}}\,\,{\text{amount}}\,\,{\text{of}}\,{\text{a}}\,\,{\text{radioactive}}\,\,{\text{substance}}\,\,{\text{present}}\,\,{\text{at}}\,{\text{time}}\,\,t,\,\,{\text{the}}\,\,{\text{the}}\,\,{\text{rate}}\,\,{\text{of}}\,\,{\text{change}}\,\,{\text{of}}\,A(t)\,\, \hfill \\
{\text{with}}\,\,{\text{respect}}\,{\text{to}}\,\,t\,\,{\text{is}}\,{\text{given}}\,{\text{b}}y\,_____. \hfill \\
\end{gathered}

Options:

A. dA=dtdA=dtdA = dt

B. A02A02\frac{{{A_0}}}{2}

C. dAdt=kA;kisaconstantofproportionalitydAdt=kA;kisaconstantofproportionality\frac{{dA}}{{dt}} = kA\,;\,\,k\,\,{\text{is}}\,\,{\text{a}}\,\,{\text{constant}}\,\,{\text{of}}\,\,{\text{proportionality}}

D. A0etA0et{A_0}\,{e^t}

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Quiz 2 - MTH401 - Differential Equations (Quiz 2)

Question:

The differential operator D transforms a _____ function into another function.

Options:

A. continuous

B. differentiable

C. discontinuous

D. integrable

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Quiz 3 - MTH401 - Differential Equations (Quiz 2)

Question:

Considerthenon - homogeneouslineardifferentialequation,any(n)+an" 1y(n" 1)+.
g(x)=e3xsin4x,thentheassumedparticularsolutionypcouldbe_____.Considerthenon - homogeneouslineardifferen
tialequation,any(n)+an" 1y(n" 1)+...+a1y2+a0y=g(x). Iftheinputfunction,g(x)=e3
utionypcouldbe_____.\begin{gathered}
{\text{Consider}}\,\,{\text{the}}\,\,{\text{non -
homogeneous}}\,\,{\text{linear}}\,\,{\text{differential}}\,\,{\text{equation,}}\,\,{a_n}{y^{(n)}} + {a_{n - 1}}{y^{(n - 1)}} + ...
+ {a_1}y' + {a_0}y = g(x).\,\, \hfill \\\ If\,\,{\text{the}}\,\,{\text{input}}\,\,{\text{function,}}\,\,g(x) = {e^{3x}}\sin 4x,\,\,{\text{
then}}\,\,{\text{the}}\,\,{\text{assumed}}\,\,{\text{particular}}\,\,{\text{solution}}\,\,{y_p}\,\,{\text{could}}\,\,{\text{be}}\,\,{\te
xt{_____}}{\text{.}}\, \hfill \\\\
\end{gathered}

Options:

A. (Ax2+Bx+C)e5x(Ax2+Bx+C)e5x(A{x^2} + Bx + C){e^{5x}}

B. (Ax+B)e5x(Ax+B)e5x(Ax + B){e^{5x}}

C. Ae5xAe5xA{e^{5x}}

D. Ae3xcos4x+Be3xsin4xAe3xcosa4x+Be3xsina4xA{e^{3x}}\cos 4x + B{e^{3x

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Quiz 4 - MTH401 - Differential Equations (Quiz 2)

Question:

A set of n functions {f1(x),f2(x),f3(x),.....,fn(x)}{f1(x),f2(x),f3(x),.....,fn(x)}\left\{
{{f_1}(x),{f_2}(x),{f_3}(x),.....,{f_n}(x)} \right\} is _____ if at least one of them can be expressed as a linear combination of the remaining.

Options:

A. linear dependent

B. linear independent

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Quiz 5 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

While using Jacobi method for the matrix
A =#¡#£#¢ 1 1 / 4 1 / 2 1 / 4 1 / 3 1 / 4 1 / 2 1 / 4 1 / 5#¤#¦#¥ A = [1 1 / 4 1 / 2 1 / 4 1 / 3 1 / 4 1 / 2 1 / 4 1 / 5] A = \left[
1&{1/4}&{1/2}\\{1/4}&{1/3}&{1/4}\\{1/2}&{1/4}&{1/5}
\end{array}} \right]
and ‘theta ¸ =0.4480, the orthogonal matrix S1 will be given by

Options:

A. S 1 =#¡#£#¢ c o s 0 . 4 4 8 0 0 0 0" s i n 0 . 4 4 8 0 0 0 0 1#¤#¦#¥ S 1 = [c o s a 0 . 4 4 8 0 0 0 0" s i n a 0 . 4 4 8 0 0
{\begin{array}{*{20}{c}}
{\cos 0.4480}&0&0\\0&{ - \sin 0.4480}&0\\0&0&1
\end{array}} \right]

B. S 1 =#¡#£#¢ 1 0 0 0 c o s 0 . 4 4 8 0 s i n 0 . 4 4 8 0 0" s i n 0 . 4 4 8 0 c o s 0 . 4 4 8 0#¤#¦#¥ S 1 = [1 0 0 0 c o s a 0
o s a 0 . 4 4 8 0] { S _ 1 } = \left[{\begin{array}{*{20}{c}}
1&0&0\\0&{\cos 0.4480}&{ - \sin 0.4480}\\0&{\sin 0.4480}&{\cos 0.4480}
\end{array}} \right]

C. S 1 =#¡#£#¢ c o s 0 . 4 4 8 0 0 s i n 0 . 4 4 8 0 0 1 0" s i n 0 . 4 4 8 0 0 c o s 0 . 4 4 8 0#¤#¦#¥ S 1 = [c o s a 0 . 4 4 8
o s a 0 . 4 4 8 0] { S _ 1 } = \left[{\begin{array}{*{20}{c}}
{\cos 0.4480}&0&{ - \sin 0.4480}\\0&1&0\\{\sin 0.4480}&0&{\cos 0.4480}
\end{array}} \right]

D. S 1 =#¡#£#¢ c o s 0 . 4 4 8 0 s i n 0 . 4 4 8 0 0" s i n 0 . 4 4 8 0 c o s 0 . 4 4 8 0 0 0 0 1#¤#¦#¥ S 1 = [c o s a 0 . 4 4 8
44800001]{S_1} = \left[{\begin{array}{*{20}{c}}
{\cos 0.4480}&{ - \sin 0.4480}&0\\{\sin 0.4480}&{\cos 0.4480}&0\\0&0&1
\end{array}} \right]

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Quiz 6 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

Power method is applicable if the eigen vectors corresponding to eigen values are linearly _____.

Options:

A. dependent

B. independent

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Quiz 7 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

Whichofthefollowingisaforwarddifferencetableforthegivenvaluesofxandy?xy0.10.0030.70.2481.30.697Whichofthefollowingisaforwarddifferencetableforthegivenvaluesofxandy?x0.10.71.3y0.0030.2480.697\begin{array}{l}
Which\,of\,the\,following\,is\,a\,forward\,difference\,table\,for\,the\,given\,values\,of\,x\,and\,y?\\\begin{array}{*{20}{c}}
x&{0.1}&{0.7}&{1.3}\\y&{0.003}&{0.248}&{0.697}
\end{array}
\end{array}

Options:

A. x0.10.71.3y0.0030.2480.697”y0.2040.449”2y0.2450xy”y”2y0.10.0030.2040.
gin{array}{*{20}{c}}
x&y&{\Delta y}&{{\Delta ^2}y}\\{0.1}&{0.003}&{0.204}&{0.245}\\{0.7}&{0.248}&{0.449}&{}\\{1.3}&{0.697}&{}&{}
\end{array}

B. x0.10.71.3y0.0030.2480.697”y0.2450.449”2y0.204xy”y”2y0.10.0030.2450.
gin{array}{*{20}{c}}
x&y&{\Delta y}&{{\Delta ^2}y}\\{0.1}&{0.003}&{0.245}&{0.204}\\{0.7}&{0.248}&{0.449}&{}\\{1.3}&{0.697}&{}&{}
\end{array}

C. x0.10.71.3y0.0030.2480.697”y0.4490.245”2y0.204xy”y”2y0.10.0030.4490.
gin{array}{*{20}{c}}
x&y&{\Delta y}&{{\Delta ^2}y}\\{0.1}&{0.003}&{0.449}&{0.204}\\{0.7}&{0.248}&{0.245}&{}\\{1.3}&{0.697}&{}&{}
\end{array}

D. x0.10.71.3y0.0030.2480.697”y0.2510.945”2y0.1196xy”y”2y0.10.0030.2510
begin{array}{*{20}{c}}
x&y&{\Delta y}&{{\Delta ^2}y}\\{0.1}&{0.003}&{0.251}&{0.1196}\\{0.7}&{0.248}&{0.945}&{}\\{1.3}&{0.697}&{}&{}
\end{array}

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Quiz 8 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

While using Jacobi method for the matrix
A =#¡#£#¢ 2" 10" 120002#¤#¦#¥ A = [2" 10" 120002] A = \left[{\begin{array}{*{20}{c}}
2&{ - 1}&0\\{ - 1}&2&0\\0&0&2
\end{array}} \right]
and 'theta ¸ =pi/4', the orthogonal matrix S1

Options:

A. S1=#¡#£#¢#¢cosÀ40sinÀ4010" sinÀ40cosÀ4#¤#¦#¥#¥S1=[cosaÀ40" sinaÀ4010sinaÀ40cosaÀ4]{S_1} = \left[{\begin{array}{*{20}{c}}
{\cos \frac{\pi }{4}}&0&{ - \sin \frac{\pi }{4}}\\0&1&0\\{\sin \frac{\pi }{4}}&0&{\cos \frac{\pi }{4}}
\end{array}} \right]

B. S1=#¡#£#¢#¢cosÀ4sinÀ40" sinÀ4cosÀ40001#¤#¦#¥#¥S1=[cosaÀ4" sinaÀ40sinaÀ4cosaÀ40001]{S_1} = \left[{\begin{array}{*{20}{c}}
{\cos \frac{\pi }{4}}&{ - \sin \frac{\pi }{4}}&0\\{\sin \frac{\pi }{4}}&{\cos \frac{\pi }{4}}&0\\0&0&1
\end{array}} \right]

C. S1=#¡#£#¢#¢1000cosÀ4sinÀ40" sinÀ4cosÀ4#¤#¦#¥#¥S1=[1000cosaÀ4" sinaÀ40sinaÀ4cosaÀ4]{S_1} = \left[{\begin{array}{*{20}{c}}
1&0&0\\0&{\cos \frac{\pi }{4}}&{ - \sin \frac{\pi }{4}}\\0&{\sin \frac{\pi }{4}}&{\cos \frac{\pi }{4}}
\end{array}} \right]

D. S1=#¡#£#¢#¢cosÀ4000" sinÀ40001#¤#¦#¥#¥S1=[cosaÀ4000" sinaÀ40001]{S_1} = \left[{\begin{array}{*{20}{c}}
{\cos \frac{\pi }{4}}&0&0\\0&{\cos \frac{\pi }{4}}&{ - \sin \frac{\pi }{4}}\\0&0&1
\end{array}} \right]

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Quiz 9 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

The Jacobi iteration converges, if A is strictly diagonally dominant.

Options:

A. FALSE

B. TRUE

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Quiz 10 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

Which of the following is equivalent form of the system of equations in matrix form; AX=B ?

Options:

A. X =(Inverse of A)B

B. X = B(Inverse of A)

C. XA = B

D. BX = A

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